

update on-CPU 콜체인 — config 병렬 PoC (2026-06-01)

cub_server testdb 단일 프로세스 on-CPU 캡처 (-F 999, dwarf call-graph) 에서 locator_update_force 를 root 로 한 콜체인. **funcset 제한 없이** script 에 등장한 모든 심볼이 후보 (커널 [unknown] 프레임만 제외).

root 선정 사유: 이 캡처는 복제 적용(replication apply) 경로라 클라이언트 SQL UPDATE 엔트리 qexec_execute_update 는 거의 부재(68 frame). 실제 hot path 인
slocator_repl_force → xlocator_repl_force → **locator_update_force** → heap_update_logical 의 write 진입점 locator_update_force 를 root 로 잡았다. (master-update-v2 의 qexec_execute_update 함수셋과는 경로가 달라 그대로는 빈 트리가 됨)

캡처 정보

항목	값
디렉토리	20260601-203730-cubserver-oncpu-config-parallel-poc-update/
전체 sample	126,337 (symbol 파싱된 stack)
locator_update_force 포함 sample	29,001 (전체의 22.96%) = 콜트리 분모
콜체인 함수 수	401개 (root 경로 inclusive)
진입 경로	xlocator_repl_force → locator_update_force (복제 적용)

※ 노드 % = root 포함 sample 대비 inclusive 비율 (insert MD 와 동일 규칙).

설정

항목	값
double_write_buffer_size	0
data_buffer_size	5G
log_buffer_size	5G
log_volume_size	1G
checkpoint_interval	30min
csql>	checkpoint 수행
addvoldb	100G
addvoldb	temp

상위 inclusive 함수 (top 24 / 401)

% of root	samples	함수	의미
100.00	29,001	locator_update_force	update force (root)
71.58	20,759	heap_update_logical	heap update
47.34	13,729	heap_log_update_physical	heap 물리 로깅
45.46	13,183	log_append_undoredo_recdes	undo/redo 로그 append (recdes)
45.27	13,129	log_append_undoredo_recdes2	“
45.15	13,094	log_append_undoredo_crumbs	undo/redo 로그 append (crumbs)
39.15	11,353	prior_lsa_alloc_and_copy_crumbs	prior LSA 레코드 생성

% of root	samples	함수	의미
31.68	9,188	prior_lsa_gen_undoredo_record_from_crumbs	prior LSA 레코드 조립
23.34	6,769	log_diff	redo diff 계산
23.02	6,675	locator_lock_and_get_object_with_evaluation	대상 row lock+get
19.54	5,666	heap_update_home	home 레코드 갱신
19.36	5,614	locator_lock_and_get_object_internal	lock+get 내부
17.70	5,132	__memmove_evex_unaligned_erms	memmove
15.11	4,382	spage_update	slotted page update
13.01	3,773	spage_update_record_after_compact	compact 후 갱신
11.83	3,430	lock_object	오브젝트 lock
11.18	3,241	lock_internal_perform_lock_object	lock 수행
8.49	2,462	spage_compact	페이지 compaction
8.47	2,455	lock_escalate_if_needed	lock escalation
8.31	2,410	lock_remove_all_inst_locks	inst lock 해제
7.27	2,108	log_zip	로그 압축
6.96	2,018	heap_get_last_version	최신 버전 조회
6.52	1,892	spage_get_record_data	레코드 read
6.01	1,743	heap_get_record_data_when_all_ready	레코드 read

전체 401개 함수는 아래 §"전체 함수 inclusive 표" 참고.

관찰 (update) + insert 대비

- update 무게중심은 **로그 경로**:
 heap_update_logical (71.6%) → heap_log_update_physical (47.3%) → log_append_undoredo_recdes (2) (45%) → log_append_undoredo_crumbs (45%) → prior_lsa_alloc_and_copy_crumbs (39%).
 insert(log_append_undoredo_crumbs 27.8%)보다 **로그 append 비중이 확연히 큼**.
- log_diff (23.3%) 가 update 에서만 큼 — 갱신 전후 diff 로 redo 를 줄이는 경로. insert 엔 없음.
- 대상 행 확보: locator_lock_and_get_object_with_evaluation (23%) +
 lock_escalate_if_needed/lock_remove_all_inst_locks (~8%) → update 는 기존 **row lock+escalation** 부담이 별도로 존재.
- 페이지 갱신: spage_update → spage_update_record_after_compact → spage_compact (가변길이 레코드 compaction) 경로가 insert 의 spage_insert_at 자리를 대체.
- 공통: prior_lsa_* + malloc 체인은 insert·update 양쪽 모두 두꺼움 (로그 레코드 생성 비용).

콜트리 (locator_update_force = 100%, 노드컷 0.1%, MIN_EDGE 30)

```
# locator_update_force 콜체인 (ROOT 포함 sample 만, ROOT = 100%)
# root_samples = 29001 (total_samples = 126337)
# 노드 컷 = 0.1% of root, MIN_EDGE = 30
```

```
[100.00%] locator_update_force
├─ [ 71.58%] heap_update_logical
│   └─ [ 47.34%] heap_log_update_physical
│       └─ [ 45.46%] log_append_undoredo_recdes
│           └─ [ 45.27%] log_append_undoredo_recdes2
```

```

└─ [ 45.15%] log_append_undo redo_crumbs
    └─ [ 39.15%] prior_lsa_alloc_and_copy_crumbs
        └─ [ 31.68%] prior_lsa_gen_undo redo_record_from_crumbs
            └─ [ 23.34%] log_diff
                └─ [ 7.27%] log_zip
                    └─ [ 5.69%] LZ4_resetStream_fast
                        └─ [ 5.24%] __memset_evex_unaligned_erms
                            └─ [ 0.33%] __tls_get_addr
                                └─ [ 0.14%] log_zip_realloc_if_needed
                                    └─ [ 0.11%] LZ4_compress_fast_extState@plt
                                        └─ [ 0.11%] LZ4_compressBound
                                            └─ [ 0.11%] log_zip_realloc_if_needed@plt
                                                └─ [ 17.70%] __memmove_evex_unaligned_erms
                                                    └─ [ 4.97%] cub_alloc
                                                        └─ [ 5.40%] __GI___libc_malloc
                                                            └─ [ 3.77%] _int_malloc
                                                                └─ [ 1.01%] unlink_chunk.isra.2
                                                                    └─ [ 0.36%] LOG_FIND_CURRENT_TDES
                                                                        └─ [ 1.34%] LOG_FIND_TDES
                                                                            └─ [ 7.27%] log_zip
                                                                                └─ [ 5.69%] LZ4_resetStream_fast
                                                                                    └─ [ 5.24%] __memset_evex_unaligned_erms
                                                                                        └─ [ 0.33%] __tls_get_addr
                                                                                            └─ [ 0.14%] log_zip_realloc_if_needed
                                                                                                └─ [ 0.11%] LZ4_compress_fast_extState@plt
                                                                                                    └─ [ 0.11%] LZ4_compressBound
                                                                                                        └─ [ 0.11%] log_zip_realloc_if_needed@plt
                                                                                                            └─ [ 4.97%] cub_alloc
                                                                                                                └─ [ 5.40%] __GI___libc_malloc
                                                                                                                    └─ [ 3.77%] _int_malloc
                                                                                                                        └─ [ 1.01%] unlink_chunk.isra.2
                                                                                                                            └─ [ 1.20%] prior_lsa_copy_undo_data_to_node
                                                                                                                                └─ [ 4.97%] cub_alloc
                                                                                                                                    └─ [ 5.40%] __GI___libc_malloc
                                                                                                                                        └─ [ 3.77%] _int_malloc
                                                                                                                                            └─ [ 1.01%] unlink_chunk.isra.2
                                                                                                                                                └─ [ 17.70%] __memmove_evex_unaligned_erms
                                                                                                                                                    └─ [ 1.19%] prior_lsa_copy_redo_data_to_node
                                                                                                                                                        └─ [ 4.97%] cub_alloc
                                                                                                                                                            └─ [ 5.40%] __GI___libc_malloc
                                                                                                                                                                └─ [ 3.77%] _int_malloc
                                                                                                    └─ [ 1.01%] unlink_chunk.isra.2
                                                                                                        └─ [ 0.22%] log_zip@plt
                                                                                                            └─ [ 3.49%] prior_lsa_next_record_internal
                                                                                                                └─ [ 2.59%] __GI___pthread_mutex_lock
                                                                                                                    └─ [ 0.25%] __lll_lock_wait
                                                                                                                        └─ [ 2.24%] __pthread_mutex_unlock_usercnt
                                                                                                                            └─ [ 0.25%] __lll_unlock_wake
                                                                                                                                └─ [ 0.35%] vacuum_get_log_blockid
                                                                                                                                    └─ [ 0.23%] prior_lsa_append_data
                                                                                                                                        └─ [ 0.14%] prior_lsa_start_append
                                                                                                                                            └─ [ 1.12%] pgbuf_set_lsa
                                                                                                                                                └─ [ 0.75%] fileio_set_page_lsa
                                                                                                                                                    └─ [ 0.78%] LSA_COPY
                                                                                                                                                        └─ [ 0.29%] pgbuf_set_dirty_buffer_ptr
                                                                                                                                                            └─ [ 0.27%] pgbuf_is_lsa_temporary
                                                                                                                                            └─ [ 0.28%] xdisk_get_purpose

```

```

└─ [ 0.18%] log_can_skip_undo_logging
    └─ [ 0.12%] log_tdes::is_system_worker_transaction@plt
        └─ [ 0.24%] logtb_find_client_type
            └─ [ 1.34%] LOG_FIND_TDES
└─ [ 0.14%] log_append_undoredo_recdes2@plt
└─ [ 1.18%] heap_page_update_chain_after_mvcc_op
    └─ [ 0.52%] spage_get_record
        └─ [ 0.48%] spage_find_slot
            └─ [ 0.13%] spage_is_unknown_slot
└─ [ 0.32%] heap_page_get_vacuum_status
    └─ [ 0.52%] spage_get_record
        └─ [ 0.48%] spage_find_slot
            └─ [ 0.13%] spage_is_unknown_slot
└─ [ 0.11%] log_append_undoredo_recdes@plt
└─ [ 19.54%] heap_update_home
    └─ [ 2.75%] spage_is_updatable
        └─ [ 4.16%] spage_check_updatable
            └─ [ 3.75%] spage_has_enough_total_space
                └─ [ 2.97%] spage_get_total_saved_spaces
                    └─ [ 2.74%] spage_get_saved_spaces
                        └─ [ 2.24%] lf_hash_find
                            └─ [ 1.35%] lf_list_find
                                └─ [ 0.19%] lf_tran_end@plt
                                    └─ [ 0.53%] lf_tran_start
                                        └─ [ 0.17%] ATOMIC_INC_64<unsigned long, int>
                                            └─ [ 0.30%] lf_callback_vpid_hash
└─ [ 0.52%] logtb_find_tranid
    └─ [ 1.34%] LOG_FIND_TDES
└─ [ 0.24%] logtb_find_current_tranid@plt
└─ [ 0.18%] logtb_is_active
    └─ [ 0.19%] log_is_in_crash_recovery
└─ [ 0.48%] spage_find_slot
    └─ [ 0.13%] spage_is_unknown_slot
└─ [ 0.68%] heap_update_set_prev_version
    └─ [ 0.22%] spage_get_record@plt
        └─ [ 0.16%] or_mvcc_set_log_lsa_to_record
└─ [ 0.25%] heap_update_physical
    └─ [ 0.23%] spage_get_record_type
└─ [ 0.11%] spage_is_updatable@plt
└─ [ 15.11%] spage_update
    └─ [ 13.01%] spage_update_record_after_compact
        └─ [ 8.49%] spage_compact
            └─ [ 17.70%] __memmove_evex_unaligned_erms
                └─ [ 0.86%] cub_calloc
                    └─ [ 0.79%] __calloc
                        └─ [ 3.77%] _int_malloc
                            └─ [ 1.01%] unlink_chunk.isra.2
└─ [ 0.71%] __GI___qsort_r
    └─ [ 0.57%] msort_with_tmp.part.0
└─ [ 17.70%] __memmove_evex_unaligned_erms
└─ [ 4.16%] spage_check_updatable
    └─ [ 3.75%] spage_has_enough_total_space
        └─ [ 2.97%] spage_get_total_saved_spaces
            └─ [ 2.74%] spage_get_saved_spaces
                └─ [ 2.24%] lf_hash_find
                    └─ [ 1.35%] lf_list_find
                        └─ [ 0.19%] lf_tran_end@plt

```

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```
# orphan (root 경로에 등장하나 트리 도달 불가, >= 0.1%):
- [ 0.24%] perfmon_is_perf_tracking
```

```

- [ 0.18%] oid_check_cached_class_oid
- [ 0.16%] prn_get_integer_value
- [ 0.16%] __GI___libc_free
- [ 0.15%] thread_get_thread_entry_info
- [ 0.14%] pgbuf_get_vpid_ptr@plt
- [ 0.13%] pgbuf_find_thrd_holder
- [ 0.13%] perfmon_inc_stat
- [ 0.13%] perfmon_add_stat
- [ 0.13%] std::_Function_base::_Base_manager<std::_Bind<void (cubmem::single_block_allocator::*(cubmem::single_block_allocator*, std::_Placeholder<1>, std::_Placeholder<2>))(cubmem::block&, unsigned long)>::_M_manager
- [ 0.13%] pthread_mutex_lock@plt
- [ 0.12%] cubthread::get_entry
- [ 0.11%] lock_conv
- [ 0.11%] xdisk_get_purpose@plt
- [ 0.11%] pgbuf_is_temp_lsa
- [ 0.11%] memcpy@plt

```

전체 함수 inclusive 표 (401개 전부, 컷오프 없음)

```

# locator_update_force 경로 inclusive 함수 표 - 전체 401개 (컷오프 없음)
# denom = root_samples = 29001

```

%of_root	samples	func
100.000	29001	locator_update_force
71.580	20759	heap_update_logical
47.340	13729	heap_log_update_physical
45.457	13183	log_append_undoredo_recdes
45.271	13129	log_append_undoredo_recdes2
45.150	13094	log_append_undoredo_crums
39.147	11353	prior_lsa_alloc_and_copy_crums
31.682	9188	prior_lsa_gen_undoredo_record_from_crums
23.341	6769	log_diff
23.016	6675	locator_lock_and_get_object_with_evaluation
19.537	5666	heap_update_home
19.358	5614	locator_lock_and_get_object_internal
17.696	5132	__memmove_evex_unaligned_erms
15.110	4382	spage_update
13.010	3773	spage_update_record_after_compact
11.827	3430	lock_object
11.175	3241	lock_internal_perform_lock_object
8.489	2462	spage_compact
8.465	2455	lock_escalate_if_needed
8.310	2410	lock_remove_all_inst_locks
7.269	2108	log_zip
6.958	2018	heap_get_last_version
6.524	1892	spage_get_record_data
6.010	1743	heap_get_record_data_when_all_ready
5.693	1651	LZ4_resetStream_fast
5.400	1566	__GI___libc_malloc
5.245	1521	__memset_evex_unaligned_erms
4.969	1441	cub_alloc
4.158	1206	spage_check_updatable
4.138	1200	lock_remove_resource
4.107	1191	lf_hash_delete_already_locked
3.914	1135	lf_list_delete


```

3.765    1092  _int_malloc
3.748    1087  spage_has_enough_total_space
3.600    1044  locator_check_foreign_key
3.493    1013  prior_lsa_next_record_internal
3.262     946  lock_internal_perform_unlock_object
2.972     862  spage_get_total_saved_spaces
2.938     852  pgbuf_unfix
2.896     840  lf_freelist_retire
2.824     819  lock_res_key_compare
2.752     798  spage_is_updatable
2.738     794  spage_get_saved_spaces
2.590     751  __GI___pthread_mutex_lock
2.245     651  lf_hash_find
2.241     650  __pthread_mutex_unlock_usercnt
2.217     643  lf_freelist_transport
2.145     622  _int_free
1.934     561  lock_dealloc_entry
1.931     560  cubthread::lockfree_hashmap<lk_res_key, lk_res>::find_or_insert
1.862     540  pgbuf_unlatch_bcb_upon_unfix
1.845     535  heap_scan_cache_allocate_area
1.779     516  lf_hash_insert_internal
1.572     456  heap_attrinfo_start_with_index
1.528     443  heap_clean_get_context
1.493     433  lf_list_insert_internal
1.348     391  lf_list_find
1.345     390  LOG_FIND_TDES
1.231     357  heap_sancache::reserve_area
1.203     349  prior_lsa_copy_undo_data_to_node
1.203     349  malloc_consolidate
1.186     344  prior_lsa_copy_redo_data_to_node
1.176     341  heap_page_update_chain_after_mvcc_op
1.124     326  heap_sancache::alloc_area
1.124     326  pgbuf_set_lsa
1.010     293  unlink_chunk.isra.2
0.969     281  cubmem::single_block_allocator::single_block_allocator
0.886     257  heap_classrepr_get
0.886     257  heap_attrinfo_end
0.883     256  operator new
0.855     248  cub_calloc
0.786     228  __calloc
0.779     226  LSA_COPY
0.752     218  fileio_set_page_lsa
0.714     207  __GI___qsort_r
0.686     199  heap_attrinfo_read_dbvalues
0.679     197  heap_update_set_prev_version
0.624     181  cubmem::block_allocator::block_allocator
0.597     173  heap_classrepr_free
0.586     170  cubmem::single_block_allocator::reserve
0.572     166  std::_Function_base::_Base_manager<std::_Bind<void (cubmem::single_block_allocator::*(cubmem::single_block_allocator*, std::_Placeholder<1>))(cubmem::block&)> >::_M_manager
0.569     165  msort_with_tmp.part.0
0.528     153  lf_tran_start
0.517     150  spage_get_record
0.517     150  logtb_find_tranid
0.500     145  mvcc_is_mvcc_disabled_class
0.479     139  spage_find_slot
0.455     132  heap_update_adjust_recdes_header

```

0.428	124	heap_get_mvcc_header
0.410	119	or_mvcc_get_header
0.379	110	spage_save_space
0.376	109	heap_sancache::assign_recdes_to_area
0.366	106	heap_attrvalue_read
0.359	104	LOG_FIND_CURRENT_TDES
0.355	103	lock_find_class_entry
0.348	101	vacuum_get_log_blockid
0.328	95	__tls_get_addr
0.321	93	heap_page_get_vacuum_status
0.303	88	lf_callback_vpid_hash
0.297	86	heap_create_update_context
0.293	85	pgbuf_set_dirty_buffer_ptr
0.290	84	heap_attrinfo_recache_attrepr
0.279	81	xdisk_get_purpose
0.269	78	logtb_get_current_mvccid
0.269	78	pgbuf_is_lsa_temporary
0.262	76	lock_res_key_hash
0.255	74	heap_sancache_block_allocate
0.252	73	heap_update_physical
0.252	73	__l1ll_lock_wait
0.248	72	__l1ll_unlock_wake
0.245	71	logtb_find_current_tranid@plt
0.245	71	perfmon_is_perf_tracking
0.238	69	heap_clear_operation_context
0.238	69	logtb_find_client_type
0.234	68	spage_get_record_type
0.231	67	pgbuf_wakeup_reader_writer
0.228	66	prior_lsa_append_data
0.228	66	lock_get_hash_value
0.224	65	pgbuf_lru_boost_bcb
0.224	65	spage_get_record@plt
0.224	65	set_waiter_exists
0.224	65	log_zip@plt
0.217	63	pgbuf_unlatch_thrd_holder
0.214	62	heap_attrvalue_point_fixed
0.210	61	heap_prepare_get_context
0.210	61	heap_get_record_location
0.210	61	lock_dealloc_resource
0.197	57	ATOMIC_INC_32<int, int>
0.197	57	pgbuf_ordered_unfix
0.193	56	log_is_in_crash_recovery
0.190	55	pgbuf_set_dirty
0.190	55	lf_hash_delete_internal
0.186	54	lf_tran_end@plt
0.183	53	oid_check_cached_class_oid
0.183	53	log_can_skip_undo_logging
0.176	51	logtb_is_active
0.172	50	heap_sancache_check_with_hfid
0.172	50	ATOMIC_INC_64<unsigned long, int>
0.166	48	hl_lea_alloc
0.159	46	prm_get_integer_value
0.155	45	or_mvcc_set_log_lsa_to_record
0.155	45	__GI__libc_free
0.152	44	thread_get_thread_entry_info
0.152	44	__GI__pthread_mutex_trylock
0.145	42	logtb_find_isolation

```

0.145      42 lock_delete_from_tran_hold_list
0.141      41 lock_get_class_lock
0.141      41 pgbuf_get_vpid_ptr@plt
0.141      41 lock_event_set_xasl_id_to_entry
0.138      40 prior_lsa_start_append
0.138      40 log_append_undoredo_recdes2@plt
0.138      40 log_zip_realloc_if_needed
0.134      39 pgbuf_find_thrd_holder
0.131      38 perfmon_inc_stat
0.131      38 perfmon_add_stat
0.128      37 spage_is_unknown_slot
0.128      37 std::_Function_base::_Base_manager<std::_Bind<void (cubmem::single_block_allocator::*(cubmem::single_block_allocator*, std::_Placeholder<1>, std::_Placeholder<2>))(cubmem::block&, unsigned long)>>::_M_manager
0.128      37 pthread_mutex_lock@plt
0.124      36 log_tdes::is_system_worker_transaction@plt
0.117      34 cubthread::get_entry
0.114      33 spage_is_updatable@plt
0.114      33 LZ4_compress_fast_extState@plt
0.114      33 lock_conv
0.114      33 check_supplemental_log
0.110      32 xdisk_get_purpose@plt
0.110      32 log_append_undoredo_recdes@plt
0.110      32 pgbuf_is_temp_lsa
0.110      32 heap_scan_pb_lock_and_fetch
0.107      31 LZ4_compressBound
0.107      31 log_zip_realloc_if_needed@plt
0.107      31 memcpy@plt
0.100      29 pgbuf_bcb_change_zone
0.100      29 spage_verify_header
0.100      29 log_is_in_crash_recovery@plt
0.097      28 mspace_malloc
0.097      28 memmove@plt
0.097      28 pgbuf_remove_watcher
0.093      27 ATOMIC_CAS_32<int, int, int>
0.093      27 malloc@plt
0.093      27 heap_init_get_context
0.093      27 lf_tran_start@plt
0.090      26 cub_free
0.090      26 logpb_get_memsize
0.090      26 heap_sancache::reserve_area@plt
0.090      26 log_tdes::is_system_worker_transaction
0.086      25 logtb_get_current_tran_index
0.086      25 heap_attrinfo_recache
0.086      25 db_private_alloc_release
0.086      25 lf_tran_compute_minimum_transaction_id
0.083      24 mvcc_is_mvcc_disabled_class@plt
0.083      24 logtb_find_tranid@plt
0.083      24 cubmem::single_block_allocator::get_ptr
0.083      24 __l1ll_lock_wait_private
0.079      23 pgbuf_notify_vacuum_follows
0.079      23 pgbuf_remove_from_lru_list
0.079      23 vacuum_is_mvccid_vacuumed
0.076      22 spage_compact@plt
0.076      22 spage_compare_slot_offset
0.076      22 pgbuf_get_vpid_ptr
0.076      22 heap_is_big_length@plt

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0.076      22 heap_attrinfo_clear_dbvalues
0.076      22 ATOMIC_CAS_ADDR<void>
0.076      22 log_prior_lsa_append_advance_when_doesnot_fit
0.072      21 vacuum_is_mvccid_vacuumed@plt
0.072      21 LZ4_compress_fast_extState
0.072      21 heap_prepare_object_page
0.072      21 db_private_get_heapid_from_thread
0.072      21 cubmem::single_block_allocator::get_size@plt
0.072      21 logtb_find_current_mvccid
0.069      20 logtb_find_current_tran_lsa@plt
0.069      20 pgbuf_remove_thrd_holder
0.066      19 pgbuf_lru_add_bcb_to_top
0.066      19 or_mvcc_get_repid_and_flags@plt
0.062      18 operator delete
0.062      18 db_private_alloc_release@plt
0.062      18 or_rep_id
0.062      18 logtb_find_current_tran_lsa
0.062      18 heap_attrvalue_transform_to_dbvalue
0.062      18 __l1_unlock_wake_private
0.062      18 lf_hash_find@plt
0.059      17 cubmem::single_block_allocator::reserve@plt
0.059      17 or_header_size
0.059      17 or_mvcc_set_log_lsa_to_record@plt
0.059      17 spage_max_record_size@plt
0.059      17 oid_is_serial
0.059      17 log_append_undo_redo_crumbs@plt
0.059      17 logtb_is_active@plt
0.059      17 heap_is_big_length
0.059      17 log_prior_lsa_append_align
0.059      17 spage_set_slot
0.059      17 locator_update_index@plt
0.055      16 oid_check_cached_class_oid@plt
0.055      16 mmon_is_memory_monitor_enabled
0.055      16 prm_get_bool_value
0.055      16 heap_sancache::alloc_area@plt
0.055      16 oid_is_serial@plt
0.052      15 pgbuf_get_tde_algorithm@plt
0.052      15 mspace_free
0.052      15 heap_sancache::assign_recdes_to_area@plt
0.048      14 log_append_get_zip_redo@plt
0.048      14 logtb_find_current_mvccid@plt
0.048      14 __GI___pthread_mutex_unlock
0.048      14 pgbuf_unfix@plt
0.048      14 heap_attrinfo_read_dbvalues@plt
0.048      14 LOG_PRIOR_LSA_LAST_APPEND_OFFSET@plt
0.048      14 or_rep_id@plt
0.048      14 locator_lock_and_get_object_with_evaluation@plt
0.048      14 pgbuf_get_page_ptype
0.048      14 pgbuf_bcb_update_flags
0.048      14 operator new@plt
0.048      14 spage_get_slot
0.045      13 locator_update_index
0.045      13 prior_lsa_alloc_and_copy_crumbs@plt
0.045      13 or_mvcc_get_repid_and_flags
0.045      13 pgbuf_get_holder
0.041      12 cubmem::single_block_allocator::get_ptr@plt
0.041      12 pthread_mutex_unlock@plt

```

```

0.041      12  log_diff@plt
0.041      12  __gthread_mutex_unlock
0.038      11  std::_Function_handler<void (cubmem::block&, unsigned long), void (*)(cubmem::block&,
unsigned long)>::_M_invoke
0.038      11  db_value_domain_init
0.038      11  free@plt
0.038      11  or_mvcc_get_prev_version_lsa
0.038      11  pgbuf_ordered_fix_release
0.038      11  calloc@plt
0.038      11  logtb_get_current_mvccid@plt
0.038      11  sys_alloc
0.034      10  spage_max_record_size
0.034      10  LSA_ISNULL
0.034      10  LOG_PRIOR_LSA_LAST_APPEND_OFFSET
0.034      10  pgbuf_ordered_fix_release@plt
0.034      10  pgbuf_set_lsa@plt
0.034      10  lock_check_escalate
0.034      10  heap_page_get_vacuum_status@plt
0.034      10  log_tdes::is_system_transaction
0.034      10  __bswap_32
0.034      10  memset@plt
0.034      10  __tls_get_addr@plt
0.034      10  css_internal_request_handler
0.034      10  execute_native_thread_routine
0.034      10  cubthread::worker_pool::core::worker::execute_current_task
0.034      10  xlocator_repl_force
0.034      10  slocator_repl_force
0.034      10  start_thread
0.034      10  css_server_task::execute
0.034      10  cubthread::worker_pool::core::worker::run
0.034      10  __GI__clone
0.034      10  net_server_request
0.031       9  or_mvcc_get_header@plt
0.031       9  heap_create_update_context@plt
0.031       9  LSA_SET_NULL
0.031       9  qsort@plt
0.031       9  lf_tran_end
0.031       9  pgbuf_set_dirty@plt
0.031       9  pgbuf_get_tde_algorithm
0.028       8  __gthread_mutex_lock
0.028       8  logtb_find_client_type@plt
0.028       8  logtb_find_current_isolation
0.028       8  heap_scan_cache_allocate_area@plt
0.028       8  __GI_qsort
0.028       8  heap_attrinfo_end@plt
0.024       7  prior_lsa_end_append
0.024       7  heap_get_mvcc_header@plt
0.024       7  lf_hash_find_or_insert@plt
0.024       7  vacuum_get_log_blockid@plt
0.024       7  spage_get_record_type@plt
0.024       7  heap_attrinfo_start_with_index@plt
0.024       7  pgbuf_lru_advance_victim_hint
0.024       7  pgbuf_get_page_ptype@plt
0.024       7  logtb_find_isolation@plt
0.024       7  heap_clean_get_context@plt
0.024       7  pgbuf_check_bcb_page_vpid
0.021       6  cubthread::get_entry@plt

```

```

0.021      6 or_mvcc_get_chn
0.021      6 prior_lsa_next_record@plt
0.021      6 log_tdes::is_system_transaction@plt
0.021      6 heap_prepare_get_context@plt
0.021      6 spage_update@plt
0.021      6 tp_domain_disk_size
0.021      6 logtb_get_current_tran_index@plt
0.021      6 heap_link_watchers
0.021      6 or_mvcc_get_flag
0.021      6 or_init
0.017      5 prior_lsa_next_record
0.017      5 db_private_free_release
0.017      5 log_append_get_zip_undo@plt
0.017      5 LZ4_resetStream_fast@plt
0.017      5 heap_get_last_version@plt
0.017      5 logtb_find_current_isolation@plt
0.017      5 ATOMIC_CAS_ADDR<pgbuf_bcb>
0.017      5 heap_unfix_watchers
0.017      5 pr_clear_value
0.017      5 log_append_get_zip_undo
0.017      5 log_append_get_zip_redo
0.017      5 lf_list_find@plt
0.017      5 logtb_find_current_tranid
0.017      5 LZ4_compressBound@plt
0.017      5 pgbuf_is_lsa_temporary@plt
0.014      4 logpb_get_memsized@plt
0.014      4 heap_get_record_data_when_all_ready@plt
0.014      4 pgbuf_ordered_unfix@plt
0.014      4 log_prior_lsa_append_add_align
0.014      4 mspace_malloc@plt
0.014      4 or_chn@plt
0.014      4 lock_object@plt
0.014      4 db_value_domain_init@plt
0.014      4 lf_hash_find_or_insert
0.014      4 or_header_size@plt
0.010      3 heap_scan_cache_allocate_recdes_data
0.010      3 spage_get_slot@plt
0.010      3 pr_clear_value@plt
0.010      3 hl_lea_free@plt
0.010      3 heap_attrinfo_clear_dbvalues@plt
0.010      3 hl_lea_alloc@plt
0.010      3 PGBUF_THREAD_HAS_PRIVATE_LRU
0.010      3 pr_type_from_id
0.010      3 spage_is_record_located_at_end
0.010      3 log_append_realloc_data_ptr
0.010      3 __GI_madvise
0.007      2 pr_type_from_id@plt
0.007      2 heap_classrepr_get@plt
0.007      2 pgbuf_notify_vacuum_follows@plt
0.007      2 mr_data_readval_int
0.007      2 lock_compat
0.007      2 pgbuf_is_temporary_volume
0.007      2 pgbuf_bcb_set_dirty
0.007      2 heap_init_get_context@plt
0.007      2 cub_free@plt
0.007      2 cubthread::lockfree_hashmap<lk_res_key, lk_res>::find_or_insert@plt
0.007      2 lock_create_search_key

```

```

0.007      2  cubmem::single_block_allocator::get_size
0.007      2  or_chn
0.007      2  lock_free_entry
0.007      2  heap_is_valid_oid
0.007      2  lf_freelist_retire@plt
0.003      1  or_mvcc_set_repid_and_flags@plt
0.003      1  or_mvcc_set_header
0.003      1  pgbuf_bcb_register_hit_for_lru
0.003      1  hl_lea_free
0.003      1  prior_update_header_mvcc_info
0.003      1  cubmem::block_allocator::block_allocator@plt
0.003      1  or_get_int
0.003      1  pgbuf_bcb_get_zone
0.003      1  cubmem::single_block_allocator::single_block_allocator@plt
0.003      1  pgbuf_should_move_private_to_shared
0.003      1  heap_classrepr_entry_remove_from_LRU
0.003      1  heap_classrepr_get_from_record
0.003      1  vacuum_produce_log_block_data
0.003      1  heap_classrepr_free@plt
0.003      1  heap_get_file_type
0.003      1  tp_domain_disk_size@plt
0.003      1  mspace_free@plt
0.003      1  pgbuf_lru_adjust_zone1
0.003      1  cub_realloc
0.003      1  lf_tran_compute_minimum_transaction_id@plt
0.003      1  lf_hash_delete_already_locked@plt
0.003      1  pthread_mutex_trylock@plt
0.003      1  lock_get_class_lock@plt
0.003      1  db_private_free_release@plt
0.003      1  thread_get_tran_entry
0.003      1  __GI___pthread_mutex_destroy

```

전체 콜체인 (모든 thread·서버 기능 포함, % of total)

위 \$콜트리는 locator_update_force subtree(=100%)만 본 것이고, 아래는 캡처된 전체 **sample**을 thread(comm)별로 묶은 콜체인입니다. 노드 % = 전체 **sample(126,337)** 대비. stack truncate 를 감안해 각 sample 의 최외곽 프레임을 thread 노드에 매달아 100% 를 덮습니다.

원본: analysis/calltree_full.txt, 전 함수(1,615개) 평탄 표: analysis/inclusive_total.txt, 전체 flamegraph: analysis/flame_full.svg.

thread(서버 스레드) 분포

% of total	samples	thread	역할
73.47	92,816	transaction	워커(update 실행)
12.26	15,489	connections	클라이언트 연결 처리
4.11	5,195	dwb-flush-block	double write buffer flush
3.47	4,378	vacuum	vacuum 워커
1.91	2,414	coordinator	트랜잭션 coordinator
1.69	2,137	log-flush	WAL flush
0.89	1,123	vacuum-master	vacuum 마스터
0.68	861	dwb-file-sync	dwb 파일 sync

% of total	samples	thread	역할
0.34	424	pgbuf-flush-con	page buffer flush
1.19	1,500	기타 8종	deadlock-detect, pgbuf-maintain, log-clock 등

전체 콜트리 (노드컷 0.2% of total, thread컷 0.3%, MIN_EDGE 30)

```
# 전체 on-CPU 콜체인 (모든 thread·함수, 노드 % = of total_samples=126337)
# 함수 노드 컷 0.2% / thread 컷 0.3% / MIN_EDGE 30

[100.00%] (ALL on-CPU) total_samples=126337
├ [ 73.47%] «transaction» (92816 smp)
│   ├── [ 23.21%] __GI___clone
│   │   └── [ 23.21%] start_thread
│   │       └── [ 23.21%] execute_native_thread_routine
│   │           └── [ 23.21%] cubthread::worker_pool::core::worker::run
│   │               ├── [ 20.36%] cubthread::worker_pool::core::worker::execute_current_task
│   │                   └── [ 20.11%] css_server_task::execute
│   │                       ├── [ 19.18%] css_internal_request_handler
│   │                           └── [ 18.96%] net_server_request
│   │                               ├── [ 17.78%] slocator_repl_force
│   │                                   ├── [ 15.08%] xlocator_repl_force
│   │                                       └── [ 8.69%] xbtree_find_unique
│   │                                           ├── [ 4.04%] btree_key_find_and_lock_unique
│   │                                               └── [ 4.01%] btree_key_find_and_lock_unique_of_unique
│   │                                                   ├── [ 3.84%] btree_key_lock_object
│   │                                                       └── [ 7.00%] lock_object
│   │                                                           ├── [ 6.36%] lock_internal_perform_lock_object
│   │                                                               └── [ 3.12%]
│   └── cubthread::lockfree_hashmap<lk_res_key, lk_res>::find_or_insert
│       ├── [ 3.38%] lf_hash_insert_internal
│       │   └── [ 2.90%] lf_list_insert_internal
│       │       ├── [ 2.26%] lock_res_key_compare
│       │       ├── [ 0.92%] lf_freelist_claim
│       │       ├── [ 0.35%] lf_stack_pop
│       │       └── [ 0.26%]
│       └── lf_freelist_alloc_block
│           └── [ 2.03%]
│   └── __GI___libc_malloc
│       └── [ 1.59%]
│   └── _int_malloc
│       └── [ 0.58%]
│   └── malloc_consolidate
│       └── [ 0.37%]
│   └── unlink_chunk.isra.2
│       └── [ 0.37%]
│   └── unlink_chunk.isra.2
│       └── [ 0.22%]
│   └── ATOMIC_INC_32<int, int>
│       └── [ 0.54%]
│   └── lf_freelist_transport
│       └── [ 0.47%]
│   └── lock_dealloc_entry
│       └── [ 0.79%]
│   └── _int_free
│       └── [ 0.58%]
```


[illegible]

Function	Line	Column	Value
			└ [0.26%] lf_freelist_alloc_block
			└ [2.03%] __GI___libc_malloc
			└ [1.59%] _int_malloc
			└ [0.58%]
malloc_consolidate			└ [0.37%]
unlink_chunk.isra.2			└ [0.37%]
unlink_chunk.isra.2			└ [0.22%] ATOMIC_INC_32<int, int>
			└ [0.54%] lf_freelist_transport
			└ [0.47%] lock_dealloc_entry
			└ [0.79%] _int_free
			└ [0.58%]
malloc_consolidate			└ [0.37%]
unlink_chunk.isra.2			└ [0.37%]
unlink_chunk.isra.2			└ [2.09%] __pthread_mutex_unlock_usercnt
			└ [0.28%] __l1l_unlock_wake
			└ [0.26%] lock_find_class_entry
			└ [2.05%] __GI___pthread_mutex_lock
			└ [2.09%]
__pthread_mutex_unlock_usercnt			└ [0.28%] __l1l_unlock_wake
			└ [2.05%] __GI___pthread_mutex_lock
			└ [0.28%] lock_get_class_lock
			└ [2.05%] __GI___pthread_mutex_lock
			└ [2.09%] __pthread_mutex_unlock_usercnt
			└ [0.28%] __l1l_unlock_wake
			└ [0.26%] lock_find_class_entry
			└ [2.05%] __GI___pthread_mutex_lock
			└ [2.09%] __pthread_mutex_unlock_usercnt
			└ [0.28%] __l1l_unlock_wake
			└ [3.79%] btree_search_key_and_apply_functions
			└ [1.44%] btree_advance_and_find_key
			└ [3.26%] pgbuf_fix_release
			└ [0.85%] pgbuf_set_bcb_page_vpid
			└ [0.38%] pgbuf_search_hash_chain
			└ [0.50%] __GI___pthread_mutex_trylock
			└ [0.31%] pgbuf_latch_bcb_upon_fix
			└ [2.09%] __pthread_mutex_unlock_usercnt
			└ [0.28%] __l1l_unlock_wake
			└ [0.71%] btree_get_root_with_key
			└ [3.26%] pgbuf_fix_release
			└ [0.85%] pgbuf_set_bcb_page_vpid
			└ [0.38%] pgbuf_search_hash_chain
			└ [0.50%] __GI___pthread_mutex_trylock
			└ [0.31%] pgbuf_latch_bcb_upon_fix
			└ [2.09%] __pthread_mutex_unlock_usercnt
			└ [0.28%] __l1l_unlock_wake
			└ [2.18%] pgbuf_unfix
			└ [1.23%] pgbuf_unlatch_bcb_upon_unfix
			└ [2.09%] __pthread_mutex_unlock_usercnt
			└ [0.28%] __l1l_unlock_wake
			└ [2.05%] __GI___pthread_mutex_lock

unlink_chunk.isra.2

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```

└─ [ 2.05%] __GI__pthread_mutex_lock
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
└─ [ 0.28%] __l1l_unlock_wake
└─ [ 2.88%] locator_repl_prepare_force
└─ [ 1.56%] heap_get_visible_version
└─ [ 0.90%] heap_get_visible_version_internal
└─ [ 0.92%] heap_prepare_get_context
└─ [ 1.49%] heap_get_record_data_when_all_ready
└─ [ 1.69%] spage_get_record_data
└─ [ 4.68%] __memmove_evex_unaligned_erms
└─ [ 0.57%] spage_get_record
└─ [ 0.38%] spage_find_slot
└─ [ 1.04%] heap_clean_get_context
└─ [ 2.18%] pgbuf_unfix
└─ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
└─ [ 0.28%] __l1l_unlock_wake
└─ [ 2.05%] __GI__pthread_mutex_lock
└─ [ 0.26%] pgbuf_lru_move_from_private_to_shared
└─ [ 2.05%] __GI__pthread_mutex_lock
└─ [ 0.92%] heap_get_class_repr_id
└─ [ 1.51%] heap_classrepr_get
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
└─ [ 0.28%] __l1l_unlock_wake
└─ [ 2.05%] __GI__pthread_mutex_lock
└─ [ 0.50%] __GI__pthread_mutex_trylock
└─ [ 1.05%] heap_classrepr_free
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
└─ [ 0.28%] __l1l_unlock_wake
└─ [ 2.05%] __GI__pthread_mutex_lock
└─ [ 0.30%] btree_get_pkey_btld
└─ [ 1.51%] heap_classrepr_get
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
└─ [ 0.28%] __l1l_unlock_wake
└─ [ 2.05%] __GI__pthread_mutex_lock
└─ [ 0.50%] __GI__pthread_mutex_trylock
└─ [ 1.05%] heap_classrepr_free
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
└─ [ 0.28%] __l1l_unlock_wake
└─ [ 2.05%] __GI__pthread_mutex_lock
└─ [ 1.37%] xtran_server_end_topop
└─ [ 0.62%] log_sysop_attach_to_outer
└─ [ 0.38%] log_tdes::unlock_topop
└─ [ 0.38%] cubpl::get_session
└─ [ 0.27%] session_get_pl_session
└─ [ 0.29%] thread_get_thread_entry_info
└─ [ 0.23%] cubthread::get_entry
└─ [ 0.21%] __tls_get_addr
└─ [ 0.24%] rmutex_unlock
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
└─ [ 0.28%] __l1l_unlock_wake
└─ [ 0.29%] thread_get_thread_entry_info
└─ [ 0.23%] cubthread::get_entry
└─ [ 0.21%] __tls_get_addr
└─ [ 0.34%] cuberr::context::pop_error_stack_and_destroy
└─ [ 0.23%] cuberr::context::push_error_stack
└─ [ 0.51%] operator new

```

```

└─ [ 2.03%] __GI___libc_malloc
    └─ [ 1.59%] _int_malloc
        └─ [ 0.58%] malloc_consolidate
            └─ [ 0.37%] unlink_chunk.isra.2
                └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 1.66%] cub_alloc
    └─ [ 2.03%] __GI___libc_malloc
        └─ [ 1.59%] _int_malloc
            └─ [ 0.58%] malloc_consolidate
                └─ [ 0.37%] unlink_chunk.isra.2
                    └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 0.58%] xtran_server_start_topop
    └─ [ 0.47%] log_sysop_start
        └─ [ 0.21%] log_tdes::lock_topop
            └─ [ 0.38%] cubpl::get_session
                └─ [ 0.27%] session_get_pl_session
                    └─ [ 0.29%] thread_get_thread_entry_info
                        └─ [ 0.23%] cubthread::get_entry
                            └─ [ 0.21%] __tls_get_addr
└─ [ 0.67%] rmutex_lock
    └─ [ 2.05%] __GI___pthread_mutex_lock
        └─ [ 0.29%] thread_get_thread_entry_info
            └─ [ 0.23%] cubthread::get_entry
                └─ [ 0.21%] __tls_get_addr
└─ [ 0.68%] LOG_FIND_TDES
└─ [ 0.53%] heap_get_class_info
    └─ [ 0.51%] heap_hfid_cache_get
        └─ [ 3.38%] lf_hash_insert_internal
            └─ [ 2.90%] lf_list_insert_internal
                └─ [ 2.26%] lock_res_key_compare
                    └─ [ 0.92%] lf_freelist_claim
                        └─ [ 0.35%] lf_stack_pop
                            └─ [ 0.26%] lf_freelist_alloc_block
                                └─ [ 2.03%] __GI___libc_malloc
                                    └─ [ 1.59%] _int_malloc
                                        └─ [ 0.58%] malloc_consolidate
                                            └─ [ 0.37%] unlink_chunk.isra.2
                                                └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 0.22%] ATOMIC_INC_32<int, int>
    └─ [ 0.54%] lf_freelist_transport
        └─ [ 0.47%] lock_dealloc_entry
            └─ [ 0.79%] _int_free
                └─ [ 0.58%] malloc_consolidate
                    └─ [ 0.37%] unlink_chunk.isra.2
                        └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 2.05%] __GI___pthread_mutex_lock
└─ [ 0.34%] heap_sancache_start_modify
    └─ [ 0.22%] file_get_type
        └─ [ 3.26%] pgbuf_fix_release
            └─ [ 0.85%] pgbuf_set_bcb_page_vpid
                └─ [ 0.38%] pgbuf_search_hash_chain
                    └─ [ 0.50%] __GI___pthread_mutex_trylock
                        └─ [ 0.31%] pgbuf_latch_bcb_upon_fix
                            └─ [ 2.09%] __pthread_mutex_unlock_usercnt
                                └─ [ 0.28%] __l1_unlock_wake
└─ [ 2.18%] pgbuf_unfix
    └─ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix

```

[illegible]

```
tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::prepare_page(unsigned long,
tbb::detail::d2::concurrent_queue_rep<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >&,
tbb::detail::d1::cache_aligned_allocator<tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::padded_page>,
tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::padded_page*&))::
{lambda()#1}::operator()
```



```
tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::prepare_page(unsigned long,
tbb::detail::d2::concurrent_queue_rep<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >&,
tbb::detail::d1::cache_aligned_allocator<tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::padded_page>,
tbb::detail::d2::micro_queue<cubconn::connection::worker::message,
tbb::detail::d1::cache_aligned_allocator<cubconn::connection::worker::message> >::padded_page*&))::
{lambda()#1}::operator()
```

[illegible]

```

└ [ 27.14%] log_append_undo_redo_crumbs
  └ [ 25.73%] prior_lsa_alloc_and_copy_crumbs
    └ [ 16.06%] prior_lsa_gen_undo_redo_record_from_crumbs
      └ [ 18.41%] log_zip
        └ [ 16.65%] LZ4_compress_fast_extState
          └ [ 3.90%] LZ4_read_ARCH
            └ [ 1.16%] LZ4_read32
              └ [ 0.79%] LZ4_initStream
                └ [ 1.91%] __memset_evex_unaligned_erms
                  └ [ 1.31%] LZ4_resetStream_fast
                    └ [ 1.91%] __memset_evex_unaligned_erms
                      └ [ 0.21%] __tls_get_addr
                        └ [ 1.16%] LZ4_read32
                          └ [ 3.90%] LZ4_read_ARCH
                            └ [ 5.36%] log_diff
                              └ [ 4.68%] __memmove_evex_unaligned_erms
                                └ [ 1.66%] cub_alloc
                                  └ [ 2.03%] __GI___libc_malloc
                                    └ [ 1.59%] _int_malloc
                                      └ [ 0.58%] malloc_consolidate
                                        └ [ 0.37%] unlink_chunk.isra.2
                                          └ [ 0.37%] unlink_chunk.isra.2
                                └ [ 18.41%] log_zip
                                  └ [ 16.65%] LZ4_compress_fast_extState
                                    └ [ 3.90%] LZ4_read_ARCH
                                      └ [ 1.16%] LZ4_read32
                                        └ [ 0.79%] LZ4_initStream
                                          └ [ 1.91%] __memset_evex_unaligned_erms
                                            └ [ 1.31%] LZ4_resetStream_fast
                                              └ [ 1.91%] __memset_evex_unaligned_erms
                                                └ [ 0.21%] __tls_get_addr
                                                  └ [ 1.16%] LZ4_read32
                                                    └ [ 3.90%] LZ4_read_ARCH
                                                      └ [ 1.66%] cub_alloc
                                                        └ [ 2.03%] __GI___libc_malloc
                                                          └ [ 1.59%] _int_malloc
                                                            └ [ 0.58%] malloc_consolidate
                                                              └ [ 0.37%] unlink_chunk.isra.2
                                                                └ [ 0.37%] unlink_chunk.isra.2
                                  └ [ 0.28%] prior_lsa_copy_undo_data_to_node
                                    └ [ 1.66%] cub_alloc
                                      └ [ 2.03%] __GI___libc_malloc
                                        └ [ 1.59%] _int_malloc
                                          └ [ 0.58%] malloc_consolidate
                                            └ [ 0.37%] unlink_chunk.isra.2
                                              └ [ 0.37%] unlink_chunk.isra.2
                                      └ [ 4.68%] __memmove_evex_unaligned_erms
                                    └ [ 0.27%] prior_lsa_copy_redo_data_to_node
                                      └ [ 1.66%] cub_alloc
                                        └ [ 2.03%] __GI___libc_malloc
                                          └ [ 1.59%] _int_malloc
                                            └ [ 0.58%] malloc_consolidate
                                              └ [ 0.37%] unlink_chunk.isra.2
                                                └ [ 0.37%] unlink_chunk.isra.2
                                  └ [ 0.81%] prior_lsa_next_record_internal
                                    └ [ 2.05%] __GI___pthread_mutex_lock
                                      └ [ 2.09%] __pthread_mutex_unlock_usercnt

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└─ [ 0.28%] __l1_unlock_wake
    └─ [ 0.27%] pgbuf_set_lsa
└─ [ 0.27%] heap_page_update_chain_after_mvcc_op
    └─ [ 0.57%] spage_get_record
        └─ [ 0.38%] spage_find_slot
└─ [ 4.49%] heap_update_home
    └─ [ 0.64%] spage_is_updatable
        └─ [ 0.96%] spage_check_updatable
            └─ [ 0.87%] spage_has_enough_total_space
                └─ [ 0.68%] spage_get_total_saved_spaces
                    └─ [ 0.63%] spage_get_saved_spaces
                        └─ [ 0.56%] lf_hash_find
                            └─ [ 0.31%] lf_list_find
└─ [ 0.38%] spage_find_slot
    └─ [ 0.57%] spage_get_record
└─ [ 3.48%] spage_update
    └─ [ 3.00%] spage_update_record_after_compact
        └─ [ 1.95%] spage_compact
            └─ [ 4.68%] __memmove_evex_unaligned_erms
└─ [ 4.68%] __memmove_evex_unaligned_erms
└─ [ 0.96%] spage_check_updatable
    └─ [ 0.87%] spage_has_enough_total_space
        └─ [ 0.68%] spage_get_total_saved_spaces
            └─ [ 0.63%] spage_get_saved_spaces
                └─ [ 0.56%] lf_hash_find
                    └─ [ 0.31%] lf_list_find
└─ [ 0.38%] spage_find_slot
    └─ [ 0.57%] spage_get_record
└─ [ 2.18%] pgbuf_unfix
    └─ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
        └─ [ 2.09%] __pthread_mutex_unlock_usercnt
            └─ [ 0.28%] __l1_unlock_wake
└─ [ 2.05%] __GI__pthread_mutex_lock
└─ [ 0.26%] pgbuf_lru_move_from_private_to_shared
    └─ [ 2.05%] __GI__pthread_mutex_lock
└─ [ 1.69%] spage_get_record_data
    └─ [ 4.68%] __memmove_evex_unaligned_erms
└─ [ 5.30%] locator_lock_and_get_object_with_evaluation
    └─ [ 4.47%] locator_lock_and_get_object_internal
        └─ [ 7.00%] lock_object
            └─ [ 6.36%] lock_internal_perform_lock_object
                └─ [ 3.12%] cubthread::lockfree_hashmap<lk_res_key, lk_res>::find_or_insert
                    └─ [ 3.38%] lf_hash_insert_internal
                        └─ [ 2.90%] lf_list_insert_internal
                            └─ [ 2.26%] lock_res_key_compare
                                └─ [ 0.92%] lf_freelist_claim
                                    └─ [ 0.35%] lf_stack_pop
                                        └─ [ 0.26%] lf_freelist_alloc_block
                                            └─ [ 2.03%] __GI__libc_malloc
                                                └─ [ 1.59%] _int_malloc
                                                    └─ [ 0.58%] malloc Consolidate
                                                        └─ [ 0.37%] unlink_chunk.isra.2
                                                            └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 0.22%] ATOMIC_INC_32<int, int>
└─ [ 0.54%] lf_freelist_transport
    └─ [ 0.47%] lock_dealloc_entry
        └─ [ 0.79%] _int free

```

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└─ [ 0.58%] malloc_consolidate
    └─ [ 0.37%] unlink_chunk.isra.2
        └─ [ 0.37%] unlink_chunk.isra.2
            └─ [ 2.05%] __GI___pthread_mutex_lock
└─ [ 1.98%] lock_escalate_if_needed
    └─ [ 2.09%] __pthread_mutex_unlock_usercnt
        └─ [ 0.28%] __l1l_unlock_wake
└─ [ 1.91%] lock_remove_all_inst_locks
    └─ [ 0.95%] lock_remove_resource
        └─ [ 0.95%] lf_hash_delete_already_locked
            └─ [ 0.90%] lf_list_delete
                └─ [ 2.26%] lock_res_key_compare
                    └─ [ 0.67%] lf_freelist_retire
                        └─ [ 0.54%] lf_freelist_transport
                            └─ [ 0.47%] lock_dealloc_entry
                                └─ [ 0.79%] _int_free
                                    └─ [ 0.58%] malloc_consolidate
                                        └─ [ 0.37%] unlink_chunk.isra.2
                                            └─ [ 0.37%] unlink_chunk.isra.2
                                                └─ [ 0.22%] ATOMIC_INC_32<int, int>
└─ [ 0.76%] lock_internal_perform_unlock_object
    └─ [ 0.67%] lf_freelist_retire
        └─ [ 0.54%] lf_freelist_transport
            └─ [ 0.47%] lock_dealloc_entry
                └─ [ 0.79%] _int_free
                    └─ [ 0.58%] malloc_consolidate
                        └─ [ 0.37%] unlink_chunk.isra.2
                            └─ [ 0.37%] unlink_chunk.isra.2
                                └─ [ 0.22%] ATOMIC_INC_32<int, int>
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
    └─ [ 0.28%] __l1l_unlock_wake
        └─ [ 2.05%] __GI___pthread_mutex_lock
└─ [ 0.92%] lf_freelist_claim
    └─ [ 0.35%] lf_stack_pop
    └─ [ 0.26%] lf_freelist_alloc_block
        └─ [ 2.03%] __GI___libc_malloc
            └─ [ 1.59%] _int_malloc
                └─ [ 0.58%] malloc_consolidate
                    └─ [ 0.37%] unlink_chunk.isra.2
                        └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 0.22%] ATOMIC_INC_32<int, int>
└─ [ 0.54%] lf_freelist_transport
    └─ [ 0.47%] lock_dealloc_entry
        └─ [ 0.79%] _int_free
            └─ [ 0.58%] malloc_consolidate
                └─ [ 0.37%] unlink_chunk.isra.2
                    └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
    └─ [ 0.28%] __l1l_unlock_wake
└─ [ 0.26%] lock_find_class_entry
    └─ [ 2.05%] __GI___pthread_mutex_lock
        └─ [ 2.09%] __pthread_mutex_unlock_usercnt
            └─ [ 0.28%] __l1l_unlock_wake
└─ [ 2.05%] __GI___pthread_mutex_lock
└─ [ 0.28%] lock_get_class_lock
    └─ [ 2.05%] __GI___pthread_mutex_lock
        └─ [ 2.09%] __pthread_mutex_unlock_usercnt

```

```

└─ [ 0.28%] __l1_unlock_wake
└─ [ 0.26%] lock_find_class_entry
└─ [ 2.05%] __GI___pthread_mutex_lock
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
└─ [ 0.28%] __l1_unlock_wake
└─ [ 1.78%] heap_get_last_version
└─ [ 1.49%] heap_get_record_data_when_all_ready
└─ [ 1.69%] spage_get_record_data
└─ [ 4.68%] __memmove_evex_unaligned_erms
└─ [ 0.57%] spage_get_record
└─ [ 0.38%] spage_find_slot
└─ [ 0.92%] heap_prepare_get_context
└─ [ 0.43%] heap_scan_cache_allocate_area
└─ [ 0.29%] heap_scancache::reserve_area
└─ [ 0.27%] heap_scancache::alloc_area
└─ [ 0.23%] cubmem::single_block_allocator::single_block_allocator
└─ [ 0.51%] operator new
└─ [ 2.03%] __GI___libc_malloc
└─ [ 1.59%] _int_malloc
└─ [ 0.58%] malloc_consolidate
└─ [ 0.37%] unlink_chunk.isra.2
└─ [ 0.37%] unlink_chunk.isra.2
└─ [ 1.66%] cub_alloc
└─ [ 2.03%] __GI___libc_malloc
└─ [ 1.59%] _int_malloc
└─ [ 0.58%] malloc_consolidate
└─ [ 0.37%] unlink_chunk.isra.2
└─ [ 0.37%] unlink_chunk.isra.2
└─ [ 0.51%] operator new
└─ [ 2.03%] __GI___libc_malloc
└─ [ 1.59%] _int_malloc
└─ [ 0.58%] malloc_consolidate
└─ [ 0.37%] unlink_chunk.isra.2
└─ [ 0.37%] unlink_chunk.isra.2
└─ [ 1.66%] cub_alloc
└─ [ 2.03%] __GI___libc_malloc
└─ [ 1.59%] _int_malloc
└─ [ 0.58%] malloc_consolidate
└─ [ 0.37%] unlink_chunk.isra.2
└─ [ 0.37%] unlink_chunk.isra.2
└─ [ 1.04%] heap_clean_get_context
└─ [ 2.18%] pgbuf_unfix
└─ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
└─ [ 0.28%] __l1_unlock_wake
└─ [ 2.05%] __GI___pthread_mutex_lock
└─ [ 0.26%] pgbuf_lru_move_from_private_to_shared
└─ [ 2.05%] __GI___pthread_mutex_lock
└─ [ 0.83%] locator_check_foreign_key
└─ [ 1.37%] heap_attrinfo_start_with_index
└─ [ 1.51%] heap_classrepr_get
└─ [ 2.09%] __pthread_mutex_unlock_usercnt
└─ [ 0.28%] __l1_unlock_wake
└─ [ 2.05%] __GI___pthread_mutex_lock
└─ [ 0.50%] __GI___pthread_mutex_trylock
└─ [ 0.26%] heap_attrinfo_recache_attrepr
└─ [ 0.91%] heap_attrinfo_end

```

```

└─ [ 1.05%] heap_classrepr_free
    └─ [ 2.09%] __pthread_mutex_unlock_usercnt
        └─ [ 0.28%] __l1_unlock_wake
            └─ [ 2.05%] __GI___pthread_mutex_lock
└─ [ 0.65%] heap_attrinfo_read_dbvalues
    └─ [ 0.37%] heap_attrvalue_read
└─ [ 22.00%] heap_log_update_physical
    └─ [ 26.25%] log_append_undoredo_recdes
        └─ [ 27.13%] log_append_undoredo_recdes2
            └─ [ 27.14%] log_append_undoredo_crumbs
                └─ [ 25.73%] prior_lsa_alloc_and_copy_crumbs
                    └─ [ 16.06%] prior_lsa_gen_undoredo_record_from_crumbs
                        └─ [ 18.41%] log_zip
                            └─ [ 16.65%] LZ4_compress_fast_extState
                                └─ [ 3.90%] LZ4_read_ARCH
                                    └─ [ 1.16%] LZ4_read32
                                        └─ [ 0.79%] LZ4_initStream
                                            └─ [ 1.91%] __memset_evex_unaligned_erms
                                                └─ [ 1.31%] LZ4_resetStream_fast
                                                    └─ [ 1.91%] __memset_evex_unaligned_erms
                                                        └─ [ 0.21%] __tls_get_addr
                                                            └─ [ 1.16%] LZ4_read32
                                                                └─ [ 3.90%] LZ4_read_ARCH
                                                                    └─ [ 5.36%] log_diff
                                                                        └─ [ 4.68%] __memmove_evex_unaligned_erms
                                                                            └─ [ 1.66%] cub_alloc
                                                                                └─ [ 2.03%] __GI___libc_malloc
                                                                                    └─ [ 1.59%] _int_malloc
                                                                                        └─ [ 0.58%] malloc_consolidate
                                                                                            └─ [ 0.37%] unlink_chunk.isra.2
                                                                                                └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 18.41%] log_zip
    └─ [ 16.65%] LZ4_compress_fast_extState
        └─ [ 3.90%] LZ4_read_ARCH
            └─ [ 1.16%] LZ4_read32
                └─ [ 0.79%] LZ4_initStream
                    └─ [ 1.91%] __memset_evex_unaligned_erms
                        └─ [ 1.31%] LZ4_resetStream_fast
                            └─ [ 1.91%] __memset_evex_unaligned_erms
                                └─ [ 0.21%] __tls_get_addr
                                    └─ [ 1.16%] LZ4_read32
                                        └─ [ 3.90%] LZ4_read_ARCH
                                            └─ [ 1.66%] cub_alloc
                                                └─ [ 2.03%] __GI___libc_malloc
                                                    └─ [ 1.59%] _int_malloc
                                                        └─ [ 0.58%] malloc_consolidate
                                                            └─ [ 0.37%] unlink_chunk.isra.2
                                                                └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 0.28%] prior_lsa_copy_undo_data_to_node
    └─ [ 1.66%] cub_alloc
        └─ [ 2.03%] __GI___libc_malloc
            └─ [ 1.59%] _int_malloc
                └─ [ 0.58%] malloc_consolidate
                    └─ [ 0.37%] unlink_chunk.isra.2
                        └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 4.68%] __memmove_evex_unaligned_erms
└─ [ 0.27%] prior_lsa_copy_redo_data_to_node

```

```

└ [ 1.66%] cub_alloc
  └ [ 2.03%] __GI___libc_malloc
    └ [ 1.59%] _int_malloc
      └ [ 0.58%] malloc_consolidate
        └ [ 0.37%] unlink_chunk.isra.2
          └ [ 0.37%] unlink_chunk.isra.2
└ [ 0.81%] prior_lsa_next_record_internal
  └ [ 2.05%] __GI___pthread_mutex_lock
    └ [ 2.09%] __pthread_mutex_unlock_usercnt
      └ [ 0.28%] __l1l_unlock_wake
└ [ 0.27%] pgbuf_set_lsa
└ [ 0.27%] heap_page_update_chain_after_mvcc_op
  └ [ 0.57%] spage_get_record
    └ [ 0.38%] spage_find_slot
└ [ 26.25%] log_append_undoredo_recdes
  └ [ 27.13%] log_append_undoredo_recdes2
    └ [ 27.14%] log_append_undoredo_crumbs
      └ [ 25.73%] prior_lsa_alloc_and_copy_crumbs
        └ [ 16.06%] prior_lsa_gen_undoredo_record_from_crumbs
          └ [ 18.41%] log_zip
            └ [ 16.65%] LZ4_compress_fast_extState
              └ [ 3.90%] LZ4_read_ARCH
                └ [ 1.16%] LZ4_read32
                  └ [ 0.79%] LZ4_initStream
                    └ [ 1.91%] __memset_evex_unaligned_erms
                      └ [ 1.31%] LZ4_resetStream_fast
                        └ [ 1.91%] __memset_evex_unaligned_erms
                          └ [ 0.21%] __tls_get_addr
                            └ [ 1.16%] LZ4_read32
                              └ [ 3.90%] LZ4_read_ARCH
                                └ [ 5.36%] log_diff
                                  └ [ 4.68%] __memmove_evex_unaligned_erms
                                    └ [ 1.66%] cub_alloc
                                      └ [ 2.03%] __GI___libc_malloc
                                        └ [ 1.59%] _int_malloc
                                          └ [ 0.58%] malloc_consolidate
                                            └ [ 0.37%] unlink_chunk.isra.2
                                              └ [ 0.37%] unlink_chunk.isra.2
└ [ 18.41%] log_zip
  └ [ 16.65%] LZ4_compress_fast_extState
    └ [ 3.90%] LZ4_read_ARCH
      └ [ 1.16%] LZ4_read32
        └ [ 0.79%] LZ4_initStream
          └ [ 1.91%] __memset_evex_unaligned_erms
            └ [ 1.31%] LZ4_resetStream_fast
              └ [ 1.91%] __memset_evex_unaligned_erms
                └ [ 0.21%] __tls_get_addr
                  └ [ 1.16%] LZ4_read32
                    └ [ 3.90%] LZ4_read_ARCH
└ [ 1.66%] cub_alloc
  └ [ 2.03%] __GI___libc_malloc
    └ [ 1.59%] _int_malloc
      └ [ 0.58%] malloc_consolidate
        └ [ 0.37%] unlink_chunk.isra.2
          └ [ 0.37%] unlink_chunk.isra.2
└ [ 0.28%] prior_lsa_copy_undo_data_to_node
└ [ 1.66%] cub_alloc

```



```

└─ [ 2.03%] __GI___libc_malloc
    └─ [ 1.59%] _int_malloc
        └─ [ 0.58%] malloc_consolidate
            └─ [ 0.37%] unlink_chunk.isra.2
                └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 4.68%] __memmove_evex_unaligned_erms
└─ [ 0.27%] prior_lsa_copy_redo_data_to_node
    └─ [ 1.66%] cub_alloc
        └─ [ 2.03%] __GI___libc_malloc
            └─ [ 1.59%] _int_malloc
                └─ [ 0.58%] malloc_consolidate
                    └─ [ 0.37%] unlink_chunk.isra.2
                        └─ [ 0.37%] unlink_chunk.isra.2
└─ [ 0.81%] prior_lsa_next_record_internal
    └─ [ 2.05%] __GI___pthread_mutex_lock
        └─ [ 2.09%] __pthread_mutex_unlock_usercnt
            └─ [ 0.28%] __lll_unlock_wake
└─ [ 0.27%] pgbuf_set_lsa
└─ [ 4.20%] locator_update_index
    └─ [ 1.11%] heap_get_class_name_alloc_if_diff
        └─ [ 0.36%] heap_sancache_end
            └─ [ 0.47%] heap_sancache_quick_end
                └─ [ 2.18%] pgbuf_unfix
                    └─ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
                        └─ [ 2.09%] __pthread_mutex_unlock_usercnt
                            └─ [ 0.28%] __lll_unlock_wake
                    └─ [ 2.05%] __GI___pthread_mutex_lock
                        └─ [ 0.26%] pgbuf_lru_move_from_private_to_shared
                            └─ [ 2.05%] __GI___pthread_mutex_lock
└─ [ 0.29%] heap_get_class_record
    └─ [ 1.78%] heap_get_last_version
        └─ [ 1.49%] heap_get_record_data_when_all_ready
            └─ [ 1.69%] spage_get_record_data
                └─ [ 4.68%] __memmove_evex_unaligned_erms
            └─ [ 0.57%] spage_get_record
                └─ [ 0.38%] spage_find_slot
└─ [ 0.92%] heap_prepare_get_context
└─ [ 1.04%] heap_clean_get_context
    └─ [ 2.18%] pgbuf_unfix
        └─ [ 1.23%] pgbuf_unlatch_bcb_upon_unfix
            └─ [ 2.09%] __pthread_mutex_unlock_usercnt
                └─ [ 0.28%] __lll_unlock_wake
        └─ [ 2.05%] __GI___pthread_mutex_lock
            └─ [ 0.26%] pgbuf_lru_move_from_private_to_shared
                └─ [ 2.05%] __GI___pthread_mutex_lock
└─ [ 1.37%] heap_attrinfo_start_with_index
    └─ [ 1.51%] heap_classrepr_get
        └─ [ 2.09%] __pthread_mutex_unlock_usercnt
            └─ [ 0.28%] __lll_unlock_wake
        └─ [ 2.05%] __GI___pthread_mutex_lock
            └─ [ 0.50%] __GI___pthread_mutex_trylock
└─ [ 0.26%] heap_attrinfo_recache_attrepr
└─ [ 0.91%] heap_attrinfo_end
    └─ [ 1.05%] heap_classrepr_free
        └─ [ 2.09%] __pthread_mutex_unlock_usercnt
            └─ [ 0.28%] __lll_unlock_wake
        └─ [ 2.05%] __GI___pthread_mutex_lock

```

```
| | | └─ [ 0.65%] heap_attrinfo_read_dbvalues  
| | |   │  
| | |   └─ [ 0.37%] heap_attrvalue_read  
| | ├── [ 0.32%] heap_attrvalue_get_key  
| | ├── [ 0.46%] btree_compare_key  
| | └── [ 0.68%] LOG_FIND_TDES  
├─ [ 2.66%] pgbuf_ordered_fix_release  
│   │  
│   └─ [ 3.26%] pgbuf_fix_release  
│       │  
│       ├── [ 0.85%] pgbuf_set_bcb_page_vpid  
│       ├── [ 0.38%] pgbuf_search_hash_chain  
│       │   │  
│       │   └─ [ 0.50%] __GI___pthread_mutex_trylock  
│       ├── [ 0.31%] pgbuf_latch_bcb_upon_fix  
│       └── [ 2.09%] __pthread_mutex_unlock_usercnt  
│           │  
│           └─ [ 0.28%] ___lwl_unlock_wake  
├─ [ 3.90%] LZ4_read_ARCH  
└─ [ 27.13%] log_append_undoredo_recdes2  
    │  
    └─ [ 27.14%] log_append_undoredo_crumbs  
        │  
        ├── [ 25.73%] prior_lsa_alloc_and_copy_crumbs  
        │   │  
        │   ├── [ 16.06%] prior_lsa_gen_undoredo_record_from_crumbs  
        │   │   │  
        │   │   └─ [ 18.41%] log_zip  
        │   │       │  
        │   │       ├── [ 16.65%] LZ4_compress_fast_extState  
        │   │       │   │  
        │   │       │   ├── [ 3.90%] LZ4_read_ARCH  
        │   │       │   │   │  
        │   │       │   │   ├── [ 1.16%] LZ4_read32  
        │   │       │   │   │   │  
        │   │       │   │   │   └─ [ 0.79%] LZ4_initStream  
        │   │       │   │   │       │  
        │   │       │   │   │       └─ [ 1.91%] __memset_evex_unaligned_erms  
        │   │       │   │   ├── [ 1.31%] LZ4_resetStream_fast  
        │   │       │   │   │   │  
        │   │       │   │   │   └─ [ 1.91%] __memset_evex_unaligned_erms  
        │   │       │   │   ├── [ 0.21%] __tls_get_addr  
        │   │       │   │   ├── [ 1.16%] LZ4_read32  
        │   │       │   │   └─ [ 3.90%] LZ4_read_ARCH  
        │   │       └─ [ 5.36%] log_diff  
        │   ├── [ 4.68%] __memmove_evex_unaligned_erms  
        │   └─ [ 1.66%] cub_alloc  
        │       │  
        │       └─ [ 2.03%] __GI__libc_malloc  
        │           │  
        │           └─ [ 1.59%] _int_malloc  
        │               │  
        │               ├── [ 0.58%] malloc_consolidate  
        │               │   │  
        │               │   └─ [ 0.37%] unlink_chunk.isra.2  
        │               └─ [ 0.37%] unlink_chunk.isra.2  
        └─ [ 18.41%] log_zip  
            │  
            ├── [ 16.65%] LZ4_compress_fast_extState  
            │   │  
            │   ├── [ 3.90%] LZ4_read_ARCH  
            │   │   │  
            │   │   ├── [ 1.16%] LZ4_read32  
            │   │   │   │  
            │   │   │   └─ [ 0.79%] LZ4_initStream  
            │   │       │  
            │   │       └─ [ 1.91%] __memset_evex_unaligned_erms  
            │   ├── [ 1.31%] LZ4_resetStream_fast  
            │   │   │  
            │   │   └─ [ 1.91%] __memset_evex_unaligned_erms  
            │   ├── [ 0.21%] __tls_get_addr  
            │   ├── [ 1.16%] LZ4_read32  
            │   └─ [ 3.90%] LZ4_read_ARCH  
            └─ [ 1.66%] cub_alloc  
                │  
                └─ [ 2.03%] __GI__libc_malloc  
                    │  
                    └─ [ 1.59%] _int_malloc  
                        │  
                        ├── [ 0.58%] malloc_consolidate  
                        │   │  
                        │   └─ [ 0.37%] unlink_chunk.isra.2  
                        └─ [ 0.37%] unlink_chunk.isra.2  
            └─ [ 0.28%] prior_lsa_copy_undo_data_to_node  
                │  
                └─ [ 1.66%] cub_alloc  
                    │  
                    └─ [ 2.03%] __GI__libc_malloc
```

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```

| | | └─ [ 2.11%] cubconn::receiver::receive
| | |   └─ [ 2.10%] __libc_recv
| | | └─ [ 0.84%] cubconn::connection::worker::handle_data_packet
| | |   └─ [ 0.60%] cubthread::worker_pool::core::execute_task
| | |     └─ [ 0.50%] cubthread::worker_pool::core::worker::assign_task
| | |       └─ [ 0.44%] std::condition_variable::notify_one
| | |         └─ [ 0.43%] __pthread_cond_signal
| | | └─ [ 0.23%] cubconn::connection::worker::handle_header_packet
| | |   └─ [ 0.51%] cubconn::receiver::release
| | |     └─ [ 0.37%] cubbase::DMRBMemoryPool::restore
| | |       └─ [ 0.21%] std::_Rb_tree<unsigned long, std::pair<unsigned long const, unsigned long>,
std::_Select1st<std::pair<unsigned long const, unsigned long> >, std::less<unsigned long>,
std::allocator<std::pair<unsigned long const, unsigned long> > >::_M_emplace_unique<unsigned long&, unsigned
long&>
| | |         └─ [ 0.24%] operator delete
| | |           └─ [ 0.26%] cub_free
| | |             └─ [ 0.34%] _int_free
| | |       └─ [ 0.49%] rmutex_lock
| | |     └─ [ 3.40%] cubsocket::epoll::wait
| | |       └─ [ 3.38%] epoll_wait
└─ [ 4.11%] «dwb-flush-block» (5195 smp)
|   └─ [ 4.05%] __GI__clone
|     └─ [ 4.05%] start_thread
|       └─ [ 4.05%] execute_native_thread_routine
|         └─ [ 4.05%] cubthread::daemon::loop_with_context
|           └─ [ 3.83%] cubthread::looper::put_to_sleep
|             └─ [ 3.67%] cubthread::waiter::wait_for
|               └─ [ 3.21%] __pthread_cond_timedwait
|                 └─ [ 0.24%] __pthread_mutex_unlock_usercnt
|                   └─ [ 0.24%] __l1l_unlock_wake
|                     └─ [ 0.24%] std::chrono::_V2::system_clock::now
|                       └─ [ 0.24%] std::chrono::_V2::system_clock::now
└─ [ 3.47%] «vacuum» (4378 smp)
|   └─ [ 2.84%] vacuum_heap_page
|     └─ [ 1.62%] vacuum_heap_prepare_record
|       └─ [ 1.49%] spage_get_record_data
|         └─ [ 1.87%] __memmove_evex_unaligned_erms
|       └─ [ 0.73%] vacuum_heap_page_log_and_reset
|         └─ [ 0.37%] heap_stats_update
|           └─ [ 0.36%] heap_stats_add_bestspace
|             └─ [ 0.33%] spage_compact
|               └─ [ 1.87%] __memmove_evex_unaligned_erms
|             └─ [ 0.25%] pgbuf_fix_release
|           └─ [ 0.45%] vacuum_process_log_block
|             └─ [ 0.31%] __GI__qsort_r
|               └─ [ 0.27%] __GI__libc_malloc
|                 └─ [ 0.25%] _int_malloc
|                   └─ [ 0.22%] malloc_consolidate
└─ [ 1.91%] «coordinator» (2414 smp)
|   └─ [ 1.87%] __GI__clone
|     └─ [ 1.87%] start_thread
|       └─ [ 1.87%] execute_native_thread_routine
|         └─ [ 1.87%] cubconn::connection::coordinator::attach
|           └─ [ 1.86%] cubconn::connection::coordinator::run
|             └─ [ 1.22%] cubsocket::epoll::wait
|               └─ [ 1.22%] epoll_wait
|             └─ [ 0.29%] cubconn::connection::coordinator::handle_message_queue

```

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|          |  └ [ 0.23%]
tbb::detail::d2::internal_try_pop_impl<tbb::detail::d2::concurrent_queue_rep<cubconn::connection::coordinator::message, tbb::detail::d1::cache_aligned_allocator<cubconn::connection::coordinator::message> >,
tbb::detail::d1::cache_aligned_allocator<tbb::detail::d2::concurrent_queue_rep<cubconn::connection::coordinator::message, tbb::detail::d1::cache_aligned_allocator<cubconn::connection::coordinator::message> > > >
|          |  └ [ 0.26%] cubconn::connection::coordinator::eventfd_clear
|          |  └ [ 0.25%] __libc_read
└ [ 1.69%] «log-flush» (2137 smp)
|   └ [ 1.18%] __GI__clone
|   |   └ [ 1.18%] start_thread
|   |   |   └ [ 1.18%] execute_native_thread_routine
|   |   |   |   └ [ 1.18%] cubthread::daemon::loop_with_context
|   |   |   |   |   └ [ 1.04%] log_flush_execute
|   |   |   |   |   |   └ [ 1.04%] logpb_flush_pages_direct
|   |   |   |   |   |   |   └ [ 0.78%] logpb_prior_lsa_append_all_list
|   |   |   |   |   |   |   |   └ [ 0.73%] logpb_append_prior_lsa_list
|   |   |   |   |   |   |   |   |   └ [ 0.50%] logpb_append_next_record
|   |   |   |   |   |   |   |   |   |   └ [ 0.21%] cub_free
|   |   |   |   |   |   |   |   |   |   |   └ [ 0.35%] logpb_append_data
|   |   |   |   |   |   |   |   |   |   |   └ [ 0.24%] logpb_flush_all_append_pages
|   |   |   |   |   |   |   |   |   |   |   └ [ 0.26%] fileio_synchronize
|   |   |   |   |   |   |   |   |   |   |   └ [ 0.26%] fdatsync
|   |   └ [ 0.25%] logpb_write_toflush_pages_to_archive
|   |   |   └ [ 0.41%] fileio_write
|   |   |   |   └ [ 0.40%] __libc_pwrite64
|   |   └ [ 0.26%] fileio_synchronize
|   |   |   └ [ 0.26%] fdatsync
└ [ 0.89%] «vacuum-master» (1123 smp)
|   └ [ 0.84%] __GI__clone
|   |   └ [ 0.83%] start_thread
|   |   |   └ [ 0.83%] execute_native_thread_routine
|   |   |   |   └ [ 0.83%] cubthread::daemon::loop_with_context
|   |   |   |   |   └ [ 0.65%] cubthread::looper::put_to_sleep
|   |   |   |   |   |   └ [ 0.63%] cubthread::waiter::wait_for
|   |   |   |   |   |   |   └ [ 0.56%] __pthread_cond_timedwait
└ [ 0.68%] «dwb-file-sync» (861 smp)
|   └ [ 0.68%] __GI__clone
|   |   └ [ 0.68%] start_thread
|   |   |   └ [ 0.68%] execute_native_thread_routine
|   |   |   |   └ [ 0.68%] cubthread::daemon::loop_with_context
|   |   |   |   |   └ [ 0.64%] cubthread::looper::put_to_sleep
|   |   |   |   |   |   └ [ 0.61%] cubthread::waiter::wait_for
|   |   |   |   |   |   |   └ [ 0.55%] __pthread_cond_timedwait
└ [ 0.34%] «pgbuf-flush-con» (424 smp)
|   └ [ 0.33%] __GI__clone
|   |   └ [ 0.33%] start_thread
|   |   |   └ [ 0.33%] execute_native_thread_routine
|   |   |   |   └ [ 0.33%] cubthread::daemon::loop_with_context
|   |   |   |   |   └ [ 0.31%] cubthread::looper::put_to_sleep
|   |   |   |   |   |   └ [ 0.31%] cubthread::waiter::wait_for
|   |   |   |   |   |   |   └ [ 0.29%] __pthread_cond_timedwait
└ [ 1.19%] «기타 thread 8종» (1500 smp, 각 <0.3%)
    0.30% deadlock-detect (376)
    0.30% pgbuf-maintain (376)
    0.22% log-clock (273)
    0.15% ha-delay-check (186)
    0.10% pgbuf-page-flus (121)

```

```
0.08% pl-monitor (104)
0.04% cub_server (55)
0.01% session-control (9)
```